HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Managed a team of **5 maintenance technicians**, overseeing daily operations, task allocation, and performance feedback.
- Implemented a **planned preventative maintenance schedule** for critical extrusion machinery, reducing unplanned downtime by **18%** over six months.
- Led **root cause investigations** for recurring equipment failures, developing and executing corrective actions that improved mean time between failures (MTBF) by **22%**.
- Provided hands-on engineering support to production teams, diagnosing and resolving complex issues on **PLC-controlled automated machinery** during high-volume manufacturing runs.
- Managed the end-to-end lifecycle of **three capital engineering projects**, from initial concept and justification through to final commissioning and operational handover.
- Ensured compliance with stringent **machinery safety standards** and coordinated statutory maintenance inspections for site utilities and production equipment.
- Containerised key operational monitoring tools with **Docker** and deployed them to an **AWS EC2** environment, improving deployment consistency and reducing setup time by **40%**.

SKILLS

Engineering & Maintenance – Plant Engineering, Maintenance Management, Reliability Engineering, Preventative Maintenance, Root Cause Analysis, Capital Projects, Commissioning, Machinery Safety, CMMS, Continuous Improvement, Lean Manufacturing

Manufacturing Systems – High-Volume Manufacturing, Extrusion Processes, Automation, PLC-Controlled Equipment, Industrial Utilities, Mechanical Systems, Electrical Systems

Leadership & Team Management – Team Leadership, Performance Management, Training & Development, Cross-functional Collaboration, Stakeholder Management

Tools & Technologies – Jira, Confluence, Splunk, Dynatrace, Jenkins, Git, Docker, AWS (EC2, S3, CloudWatch), Python Scripting, SQL

Project Management – Project Lifecycle Management, Capital Expenditure Justification, Process Trials, Production Handover

PROJECTS

Automated Extrusion Line Reliability Upgrade (HCL Technologies)

- Led a cross-functional team in upgrading a critical **Java Spring Boot**-based control system for an automated extrusion line, enhancing data logging and error reporting.
- Designed and implemented a **Python-based diagnostic tool** that analyzed system logs from **Splunk**, reducing manual troubleshooting time for common faults by approximately **30%**.
- Integrated real-time performance metrics from the upgraded line into an existing **operational dashboard**, providing management with enhanced visibility into throughput and downtime events.

Predictive Maintenance Pilot for Manufacturing Equipment (HCL Technologies)

- Developed a **Python** data pipeline to ingest sensor data (vibration, temperature) from **AWS S3** for a fleet of manufacturing machines.
- Engineered **SQL** queries against PostgreSQL to extract and transform historical maintenance logs and sensor readings for feature engineering.
- Built and evaluated a **scikit-learn** anomaly detection model on a dataset of ~55,000 sensor readings, achieving a **0.78 F1-score** in identifying precursor patterns to equipment failure.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)